

| KGCE IN ELECTRONICS ENGINEERING - MODEL CURRICULUM |                                                           |              |                 |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                      |                                                                              |                                                                                                          |
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| Programme Title : KGCE in Electronics Engineering  |                                                           |              |                 |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                      | Notional Hours: 2880                                                         |                                                                                                          |
| COURSE NAME & CODE                                 | TOPIC / MODULE                                            | THEORY (Hrs) | PRACTICAL (Hrs) | OBJECTIVE OF MODULE                                                                                                                                                                                                             | OUTCOME OF MODULE                                                                                                                                                                                                                    | METHODOLOGY                                                                  | TOOLS REQUIRED                                                                                           |
|                                                    |                                                           | 480          | 960             | YEAR 1                                                                                                                                                                                                                          |                                                                                                                                                                                                                                      |                                                                              |                                                                                                          |
| BASIC MATHEMATICS & SCIENCE (1001)                 | Module M1<br>Basic Mathematics calculations & Algebra     | 40           | 0               | OB 1.1<br>To understand principles of basic mathematics and calculation including Fraction, Ratio & Proportions, Basic Algebra                                                                                                  | Will be able to:<br>MO-1.1<br>Perform basic mathematical calculations in Fraction, Ratio & Proportions, Basic Algebra                                                                                                                | - Lecture<br>- Use of smart class rooms<br>- Use of instructional materials  | - Laptop & Projector<br>- Guideline documents                                                            |
|                                                    | Module M2<br>Mensuration and Trigonometry                 | 40           | 0               | OB 2.1<br>To understand principles of Mensuration and Trigonometry                                                                                                                                                              | MO-2.1<br>Perform basic mathematical calculations and solve sample problems related to Mensuration and Trigonometry                                                                                                                  | - Lecture<br>- Use of smart class rooms<br>- Use of instructional guidelines | - Laptop & Projector<br>- Guideline documents                                                            |
|                                                    | Module M3<br>Basic Science                                | 40           | 0               | OB 3.1<br>To understand principles of basic Science including System of units, Unit Conversion<br>Mass/weight/volume/density, Work/power/energy, Velocity/Speed, elasticity                                                     | MO-3.1<br>Understand the concepts of basic science including :<br>System of units, Unit Conversion<br>MO-3.2<br>Define - Mass/weight/volume/density, Work/power/energy, Velocity/Speed, elasticity                                   | - Lecture<br>- Use of smart class rooms<br>- Use of instructional guidelines | - Laptop & Projector<br>- Guideline documents                                                            |
|                                                    | Module M4<br>Basic Science                                | 40           | 0               | OB 4.1<br>To understand principles of basic Science including Heat, Pressure & Temperature and their applications.<br>OB 4.2<br>To Understand the concepts of Basic electricity - AC/DC/Voltage, Current , Resistance, Ohms law | MO-4.1<br>Define - Heat, Pressure & Temperature and their applications<br>MO-4.2<br>Explain - AC/DC/Voltage, Current , Resistance, Ohms law                                                                                          | - Lecture<br>- Use of smart class rooms<br>- Use of instructional guidelines | - Laptop & Projector<br>- Guideline documents                                                            |
| Engineering Drawing                                | Module M1<br>Introduction to Engineering Drawing Practice | 8            | 0               | OB 1.1<br>To understand different instruments used in engineering drawing                                                                                                                                                       | MO-1.1<br>List various instruments used in engineering drawing<br>MO-1.2<br>State uses of various drawing instruments<br>MO-1.3<br>Use various instruments to draw sample exercises                                                  | - Lecture<br>- Demonstration                                                 | - Scales, Compass, Drawing board, Clips, Mini drafter, Pencils, Drawing sheets, Stencils, Instrument box |
|                                                    |                                                           | 10           | 0               | OB 1.2<br>To understand freehand sketching, lettering and dimensioning                                                                                                                                                          | MO-1.4<br>Understand the application of freehand sketching, lettering and dimensioning, Layouting and title block<br>MO-1.5<br>List various dimensioning methods<br>MO-1.6<br>Solve problems based on different dimensioning methods | - Lecture<br>- Demonstration                                                 |                                                                                                          |
|                                                    | Module M2<br>Geometrical Drawing                          | 20           | 0               | OB 2.1<br>To understand Geometric constructions and drawings of various objects and shapes                                                                                                                                      | MO-2.1<br>Draw lines, angles, triangles, squares, polygons, threads, fasteners based on sample exercises                                                                                                                             | - Lecture<br>- Demonstration                                                 |                                                                                                          |

| COURSE NAME<br>&<br>CODE                                             | TOPIC /<br>MODULE                                                               | THEOR<br>Y (Hrs) | PRACTI<br>CAL<br>(Hrs) | OBJECTIVE OF MODULE                                                                                                                                                                                                                                                                                                                                                                   | OUTCOME OF MODULE                                                                                                                                                                                                                                                                                                                                                                                      | METHEDOLOGY                                                                        | TOOLS<br>REQUIRED                                                 |
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| (1002)                                                               | <b>Module M3</b><br>Orthographic<br>Projection                                  | 20               | 0                      | OB 3.1<br>To draw orthographic projections of various objects                                                                                                                                                                                                                                                                                                                         | MO-3.1<br>State the concept of quadrants in engineering drawing<br>MO-3.2<br>Differentiate first angle and third angle projection<br>MO-3.3<br>Prepare orthographic projection of given sample objects                                                                                                                                                                                                 | - Lecture<br>- Demonstration                                                       | - Laptop &<br>Projector                                           |
|                                                                      | <b>Module M4</b><br>Shop floor drawing                                          | 22               | 0                      | OB 4.1<br>To undestrand and draw shop floor drawings                                                                                                                                                                                                                                                                                                                                  | MO-4.1<br>State the importance of shop floor drawing in industry<br>MO-4.2<br>Prepare isometric drawings of given sample objects<br>MO-4.3<br>Prepare assembly drawing of given sample products                                                                                                                                                                                                        | - Lecture                                                                          |                                                                   |
| <b>Professional<br/>Knowledge-I<br/>(Trade Theory)</b><br><br>(1041) | <b>Module M1</b><br>Basic Electronics                                           | 50               | 0                      | OB 1.1<br>To understand the fundamentals of Electronics which includes Understand about Basic Electricity ,Understand Passive and Active Components , Understand different types of Resistors,,Capacitors and Transformers , Understand Microphone and Loud Speakers, Understand Active Components ,Describe Transistors and power devices ,Discuss IC s and Surface Mount Devices    | <b>Will be able to:</b><br><br>MO-1.1<br>Explain about the active and passive components in basic electronics                                                                                                                                                                                                                                                                                          | - Lecture<br>- Use of smart class<br>rooms<br>- Use of instructional<br>guidelines | - Laptop & Projector<br>- Electronic<br>Components                |
|                                                                      | <b>Module M2</b><br>Digital Electronics ,<br>Electronic Devices<br>and Circuits | 70               | 0                      | OB 2.1 To Explain about Number Systems , Basic Gates , Applications<br>OB 2.2 To understand the importance of Basic Electronic Circuits : , Describe Half Wave Rectifier , Describe fullWave Rectifiers (Centre tapped and Bridge), Illustrate voltage regulator and SMPS<br>Understand Amplifiers ,Assemble an Amplifier , Describe Oscillators                                      | M O 2.1 To Explain about Number Systems , Basic Gates , Applications<br>M O 2.2 To understand the importance of Basic Electronic Circuits                                                                                                                                                                                                                                                              | - Lecture<br>- Use of smart class<br>rooms<br>- Demonstration                      | - Laptop & Projector<br>- Electronic<br>Components                |
|                                                                      | <b>Module M3</b><br>Communication<br>Systems and<br>Industrial Safety           | 40               | 0                      | OB.3.1 Analog and Digital Communication<br>OB.3.2. Describe Modulation and De Modulation<br>OB.3.3. Understand Radio and Television Communication<br>OB.3.4. Describe cable T V and DTH<br>OB.3.5. Fiber to the Home - FTTH<br>OB.3.6. Describe Mobile Communication<br>OB.3.7. Apply the best customer relationship on pre and post servicing<br>OB.3.8. Apply safe working practice | M.O.3.1. Discuss Analog and Digital Communication<br>M.O.3.2. Describe Modulation and De Modulation<br>M.O.3.3. Understand Radio and Television Communication<br>M.O.3.4. Describe cable T V and DTH<br>M.O.3.5. Fiber to the Home - FTTH<br>M.O.3.6. Describe Mobile Communication<br>M.O.3.7. Apply the best customer relationship on pre and post servicing<br>M.O.3.8. Apply safe working practice | - Lecture<br>- Use of smart class<br>rooms<br>- Demonstration                      | - Laptop & Projector<br>- Electronic<br>Components and<br>Devices |

| COURSE NAME<br>&<br>CODE | TOPIC /<br>MODULE                                            | THEOR<br>Y (Hrs) | PRACTI<br>CAL<br>(Hrs) | OBJECTIVE OF MODULE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | OUTCOME OF MODULE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | METHEDOLOGY                              | TOOLS<br>REQUIRED                                           |
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|                          | <b>Module M4</b><br>ELECTRONIC<br>HOME<br>APPLIANCES         | 80               | 0                      | OB 4.1 Basic Principle, Working and Operation of LCD/LED/Smart TVs , Home Theater Sources, Analog Audio and Video Sources, Home Theater Audio Speaker, Amplifier and Power Relationships Speaker Types and Feature, Equalization Surround Sound Options Blue Tooth and wireless connectivity , LED Light system and LED Bulbs,Tubes and Emergency Lamps<br>Basic Theory of Electro magnetic Induction and principle of operation<br>Induction cook top , Electrical and electronics circuit in                                                                                                                                                                                                                                                                                                         | MO 4.1 Basic Principle, Working and Operation of LCD/LED/Smart TVs , Home Theater Sources, Analog Audio and Video Sources, Home Theater Audio Speaker, Amplifier and Power Relationships Speaker Types and Feature, Equalization Surround Sound Options Blue Tooth and wireless connectivity ,<br>MO 4.2 LED Light system and LED Bulbs,Tubes and Emergency Lamps<br>MO 4.3 Basic Theory of Electro magnetic Induction and principle of operation<br>Induction cook top , Electrical and electronics circuit in                                                                                                                                                                                                                                                                                        | - Lecture<br>- Demonstration             | - Laptop &<br>Projector                                     |
|                          | <b>Module M1</b><br>Basic Electronics<br>Workshop            | 0                | 130                    | OB 1.1<br>Make use of Multimeter , Operate Single and Dual Regulated Power Supplies , Make use of CRO/DSO ,Experiment with Function Generator , Practice neat soldering and desoldering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MO-1.1<br>Work with different devices and instruments required for the testing and study of Electronic circuits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | - Demonstration<br>- Practical Exercises | '- General tools &<br>equipments used in<br>Electronics Lab |
|                          | <b>Module M2 Digital<br/>and Electronics<br/>Circuit Lab</b> | 0                | 100                    | OB 2.1<br>Verify the Truth Tables of Basic Gates with Digital IC Trainer Kit<br>Explain Rectifiers.<br>Discuss the types of Rectifiers.<br>Understand the use of Filters<br>Assemble Half wave rectifier with and without filters<br>Assemble Full wave rectifier with and without filters<br>Assemble Bridge rectifier<br>Discuss various types of voltage regulators<br>Assemble a IC Voltage Regulator<br>Test a SMPS<br>Explain basic theory of amplification<br>Describe the working of Transistor Amplifier<br>Assemble IC Power Amplifier<br>Build up a practical Amplifier<br>PA system assembling and test with input signals<br>Understand basic principles of oscillations<br>Describe Sine wave and Squire wave Oscillators<br>Assemble an Astable Multi vibrator using Transistors and IC | MO-2.1<br>Verify the Truth Tables of Basic Gates with Digital IC Trainer Kit<br>Explain Rectifiers.<br>Discuss the types of Rectifiers.<br>Understand the use of Filters<br>Assemble Half wave rectifier with and without filters<br>Assemble Full wave rectifier with and without filters<br>Assemble Bridge rectifier<br>Discuss various types of voltage regulators<br>Assemble a IC Voltage Regulator<br>Test a SMPS<br>Explain basic theory of amplification<br>Describe the working of Transistor Amplifier<br>Assemble IC Power Amplifier<br>Build up a practical Amplifier<br>PA system assembling and test with input signals<br>Understand basic principles of oscillations<br>Describe Sine wave and Squire wave Oscillators<br>Assemble an Astable Multi vibrator using Transistors and IC | - Demonstration<br>- Practical Exercises | '- General tools &<br>equipments used in<br>Electronics Lab |

| COURSE NAME<br>&<br>CODE                                           | TOPIC /<br>MODULE                                                            | THEOR<br>Y (Hrs) | PRACTI<br>CAL<br>(Hrs) | OBJECTIVE OF MODULE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | OUTCOME OF MODULE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | METHEDOLOGY                              | TOOLS<br>REQUIRED                                     |
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| <b>Professional Skill-<br/>I (Trade Practical)<br/><br/>(1049)</b> | <b>Module M3</b><br>Communication<br>Systems and<br>Industrial Safety<br>Lab | 0                | 60                     | OB 3.1<br>Compare Analog and Digital Communication<br>Describe Modulation and types of Modulations<br>State need of modulation<br>Define demodulation<br>Know about AM/FM transmission and reception<br>Discuss cable T V and DTH<br>Know about Fiber communication<br>Name Mobile phone technologies<br>List the evolution<br>Maintain complete customer satisfaction during the complete Service cycle.<br>Effectively communicate with clients/ customers to gain commitment for Servicing.<br>Follow and maintain procedures to achieve a safe working environment in line with occupational health and safety regulations and requirements and according to site policy                                                                                                                                                                                                                                                                                                                                                              | MO-3.1<br>Compare Analog and Digital Communication<br>Describe Modulation and types of Modulations<br>State need of modulation<br>Define demodulation<br>Know about AM/FM transmission and reception<br>Discuss cable T V and DTH<br>Know about Fiber communication<br>Name Mobile phone technologies<br>List the evolution<br>Maintain complete customer satisfaction during the complete Service cycle.<br>Effectively communicate with clients/ customers to gain commitment for Servicing.<br>Follow and maintain procedures to achieve a safe working environment in line with occupational health and safety regulations and requirements and according to site policy                                                                                                                                                                                                                                                                                                                                                           | - Demonstration<br>- Practical Exercises | '- General tools & equipments used in Electronics Lab |
|                                                                    | <b>Module M4</b><br>Electronic HOME<br>APPLIANCES Lab                        | 0                | 270                    | OB 4.1<br>Discuss Basic Principle, Working and Operation of LCD/LED/Smart TVs<br>Home Theater Sources, Analog Audio and Video Sources, Digital Audio and Video Sources and Digital Media Center<br>Home Theater A/V Processing Video Basics Signal Routing Options Audio and Video Signal Types Sound and Video Connectors Analog and Digital Connections Audio Formats Video Formats<br>Home Theater Audio Speaker, Amplifier and Power Relationships Speaker Types and Features Equalization Surround Sound Options<br>Blue Tooth and wireless connectivity<br>SD/USB HDD Memory Utilization<br>Describe LED Light system<br>Assembling and Testing of LED Bulbs, Tubes and Emergency Lamps<br>Basic Theory of Electro magnetic Induction and principle of operation<br>Identify the faults in Induction cook top and rectify.<br>Dismantle and identify various parts, wiring and tracing of various controls, Electrical and electronics circuit in Induction cook top.<br>Replacing the Induction tube (coil) in Induction cook top. | MO-4.1<br>Discuss Basic Principle, Working and Operation of LCD/LED/Smart TVs<br>Home Theater Sources, Analog Audio and Video Sources, Digital Audio and Video Sources and Digital Media Center Home Theater A/V Processing Video Basics Signal Routing Options Audio and Video Signal Types Sound and Video Connectors Analog and Digital Connections Audio Formats Video Formats<br>Home Theater Audio Speaker, Amplifier and Power Relationships Speaker Types and Features Equalization Surround Sound Options<br>Blue Tooth and wireless connectivity<br>SD/USB HDD Memory Utilization<br>Describe LED Light system<br>Assembling and Testing of LED Bulbs, Tubes and Emergency Lamps<br>Basic Theory of Electro magnetic Induction and principle of operation<br>Identify the faults in Induction cook top and rectify.<br>Dismantle and identify various parts, wiring and tracing of various controls, Electrical and electronics circuit in Induction cook top.<br>Replacing the Induction tube (coil) in Induction cook top. | - Demonstration<br>- Practical Exercises | '- Electronic Home Appliances in the Lab              |

| COURSE NAME<br>&<br>CODE                                     | TOPIC /<br>MODULE                                                   | THEOR<br>Y (Hrs) | PRACTI<br>CAL<br>(Hrs) | OBJECTIVE OF MODULE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | OUTCOME OF MODULE                                                                                                                                                                                                                                                                                   | METHEDOLOGY                                                                      | TOOLS<br>REQUIRED                                                |
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|                                                              | Module M4<br>ELECTRONIC<br>HOME<br>APPLIANCES LAB<br>( by OJT mode) |                  | 400                    | OB 1.1 Deliver and setup small home appliances OB<br>1.2 Deliver and setup consumer appliances -TV, Oven ,<br>Induction Cooker and Home Theater,Emergency<br>Lamps...                                                                                                                                                                                                                                                                                                                                                                                                                | MO-1.1<br>Familiarization of Appliances to the consumers<br>MO1.2 - Provide basic training to the consumer                                                                                                                                                                                          | - Demonstration<br>- Industrial Visit-<br>Provide on call or on site<br>services | To be completed<br>outside the academic<br>hours or in vacation. |
|                                                              |                                                                     | 480              | 960                    | YEAR 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                     |                                                                                  |                                                                  |
| Professional<br>Knowledge-II<br>(Trade Theory)<br><br>(2041) | Module M1<br>Basics of DTH                                          | 50               | 0                      | OB 1.1<br>Introduction to Satellite Communication<br>Introduction to DTH , DTH Terminology and<br>Components<br>ntroduction of Tools and Basic Electronics for DTH Role<br>Basics of Electromagnetic Spectrum and Frequency<br>Bands used in Satellite Communication<br>Dish Assembly , Mount Assembly                                                                                                                                                                                                                                                                               | MO-1.1<br>Introduction to Satellite Communication<br>Introduction to DTH , DTH Terminology and Components<br>ntroduction of Tools and Basic Electronics for DTH Role<br>Basics of Electromagnetic Spectrum and Frequency Bands used<br>in Satellite Communication<br>Dish Assembly , Mount Assembly | - Lecture<br>- Use of smart class<br>rooms                                       | - Laptop &<br>Projector                                          |
|                                                              | Module M2<br>BASICS OF<br>CABLE TV                                  | 50               | 0                      | OB 2.1<br>Basics of Digital Cable Technology - Evoluton of Cable<br>and Satellite Television<br>Headend System and Distributon Network<br>MultiPlay Digital Services<br>- Access Network Architecture                                                                                                                                                                                                                                                                                                                                                                                | MO-2.1<br>Basics of Digital Cable Technology - Evoluton of Cable and<br>Satellite Television<br>Headend System and Distributon Network<br>MultiPlay Digital Services<br>- Access Network Architecture                                                                                               | - Lecture                                                                        | - Laptop &<br>Projector                                          |
|                                                              | Module M3<br>BASICS OF UPS<br>AND INVERTER                          | 65               | 0                      | OB 3.1<br>Introduction to UPS<br>Block Diagram of UPS<br>Functions of Basic elements<br>Working Principle of UPS<br>General Types of UPS in use<br>Introduction to Inverter<br>Block Diagram of Inverter and functions of each element<br>Working Principle of Inverter and its applications- Screw<br>threads, Taps & Dies : Screw thread Terminology,<br>applications<br>- Nut, Bolt, Keys, Other Fastenings<br>- Gauges<br>- Interchangeability<br>- Importance of surface finish<br>- Limits, Fits & Tolerance<br>- Precision measuring instruments<br>Linear, Angular & Surface | MO - 3.1<br>Introduction to UPS<br>Block Diagram of UPS<br>Functions of Basic elements<br>Working Principle of UPS<br>General Types of UPS in use<br>Introduction to Inverter<br>Block Diagram of Inverter and functions of each element<br>Working Principle of Inverter and its applications      | - Lecture<br>- Demonstration<br>- Use of smart class<br>rooms                    | - Laptop &<br>Projector                                          |

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|                                                                            | <b>Module M4</b><br>BASICS OF<br>SOLAR<br>ENGINEERING    | 35               | 0                      | OB 4.1<br>To understand the solar cell<br>To understand the working solar cell PN junction<br>Learn the working of Solar Panel                                 | MO-4.1<br>To understand the solar cell<br>To understand the working solar cell PN junction<br>Learn the working of Solar Panel                                                                                                                                                                                                                                 | - Lecture<br>- Demonstration<br>- Use of smart class<br>rooms                                      | - Laptop &<br>Projector |
| <b>Employability<br/>Skills &amp;<br/>Entrepreneurship<br/><br/>(2001)</b> | <b>Module M1</b><br>English &<br>Communication           | 5                | 10                     | OB 1.1<br>To understand communication and self management<br>skills<br><br>OB 1.2<br>To understand English Literacy - functional English,<br>reading & writing | MO-1.1<br>Demonstrate knowledge of various methods of communication -<br>verbal, non-verbal-visual; Greetings & self introduction, Asking<br>& responding to question, formal & informal communication<br>MO-1.2<br>Demonstration of writing sentences and paragraphs on topics<br>related to the subject, discussions on current happenings                   | - Lecture<br>- Demonstration<br>- Use of smart class<br>rooms<br>- Mock discussions,<br>Interviews | - Laptop &<br>Projector |
|                                                                            | <b>Module M2</b><br>Communication &<br>Behavioral Skills | 5                | 10                     | OB 2.1<br>To understand Behavioral skills - Personal strength<br>analysis, social responsibility, role modeling                                                | MO-2.1<br>Identify specific do's and don'ts for avoiding common body<br>language mistakes<br>MO-2.2<br>Execute time management and planning skills, Skills to crack<br>interviews<br>MO-2.3<br>Demonstration of impressive appearance and groomed<br>personality, ability to self- explore<br>MO-2.4<br>Display professionalism at the institute and workplace | - Lecture<br>- Demonstration<br>- Use of smart class<br>rooms<br>- Mock discussions,<br>Interviews | - Laptop &<br>Projector |
|                                                                            | <b>Module M3</b><br>Information<br>Technology            | 20               | 40                     | OB 3.1<br>To understand Information and communication<br>technology skills<br><br>OB 3.2<br>To be familiar with internet and its applications                  | MO-3.1<br>Understand the basics of computers, Operating system, MS-<br>Word, MS-Excel software's<br>MO-3.2<br>Create simple documents like - resume, letter writing, job<br>application etc.,<br>MO-3.3<br>Printing document, Familiar with usage of shortcuts, Creating<br>and Editing of Text, Formatting the Text.<br><del>MO-3.4</del>                     | - Lecture<br>- Demonstration<br>- Use of smart class<br>rooms                                      | - Laptop &<br>Projector |
|                                                                            | <b>Module M4</b><br>Entrepreneurship                     | 25               | 5                      | OB 4.1<br>To understand Entrepreneurial skills                                                                                                                 | MO-4.1<br>Describe the significance of entrepreneurial values and attitude.<br>MO-4.2<br>Demonstrate the knowledge of attitudinal changes required to<br>become an entrepreneur<br>MO-4.3<br>Explain the ways to set up an enterprise and different aspects<br>involved viz., legal, compliances, Marketing aspect, Budgeting,                                 | - Lecture<br>- Demonstration<br>- Use of smart class<br>rooms                                      | - Laptop &<br>Projector |



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|                                                       | <b>Module 1</b><br>DTH<br>INSTALLATION<br>AND SERVICE              | 0                | 220                    | OB 1.1 DTH Packages and Products<br>Practicals on Frequency Bands used in Satellite Communication<br>DTH Installation<br>Working and Operation of Digital Satellite Finder<br>Connectorization<br>Signal Peaking<br>HD PVR Installation and VOD connection<br>Troubleshooting<br>Demo of DTH SD/HD Box                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MO-1.1<br>DTH Packages and Products<br>Practicals on Frequency Bands used in Satellite Communication<br>DTH Installation<br>Working and Operation of Digital Satellite Finder<br>Connectorization<br>Signal Peaking<br>HD PVR Installation and VOD connection<br>Troubleshooting<br>Demo of DTH SD/HD Box                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | - Demonstration<br>- Practical Exercises | - DTH settop box<br>and Basic tools used<br>in DTH |
|                                                       | <b>Module 2</b><br>CABLE TV<br>INSTALLATION<br>AND SERVICE         | 0                | 240                    | OB 2.1<br>-Tools and Equipment<br>-Building Access Network for DOCSIS<br>-Building Outside plant of Fibre Access Network<br>Laying coaxial cables, carrying out installation and connectorization of actives and passives<br>Carrying out installation of power inserter and balancing and commissioning of amplifier<br>Pre-installation and installation process of Set Top Box (STB)<br>Cater to customer complaints by troubleshooting and maintenance of set top box<br>Understand application and Networking of Optical Devices using different OF Cables<br>Laying Optical Fiber cables, carrying out installation and connectorization of actives and passives<br>Carrying out installation of OF Cable Joints (Splicing) ,<br>Different Types of OF Closures<br>Pre-installation and installation process of Set Top Box (STB) and Optical Modem for OF cables | MO-2.1<br>-Tools and Equipment<br>-Building Access Network for DOCSIS<br>-Building Outside plant of Fibre Access Network<br>Laying coaxial cables, carrying out installation and connectorization of actives and passives<br>Carrying out installation of power inserter and balancing and commissioning of amplifier<br>Pre-installation and installation process of Set Top Box (STB)<br>Cater to customer complaints by troubleshooting and maintenance of set top box<br>Understand application and Networking of Optical Devices using different OF Cables<br>Laying Optical Fiber cables, carrying out installation and connectorization of actives and passives<br>Carrying out installation of OF Cable Joints (Splicing) ,<br>Different Types of OF Closures<br>Pre-installation and installation process of Set Top Box (STB) and Optical Modem for OF cables . | - Demonstration<br>- Practical Exercises | CABLE TV Service<br>Equipemnts and<br>Devices      |
| <b>Professional Skill-II (Trade Practical) (2049)</b> | <b>Module 3</b><br>UPS and Inverter<br>Installation and<br>Service | 0                | 250                    | OB 3.1<br>UPS Installation<br>Trouble shooting of UPS<br>Distributors of UPS<br>Installation of Inverter<br>Trouble shooting of Inverter<br>Distributors of Inverter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MO-3.1<br>UPS Installation<br>Trouble shooting of UPS<br>Distributors of UPS<br>Installation of Inverter<br>Trouble shooting of Inverter<br>Distributors of Inverter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | - Demonstration<br>- Practical Exercises | UPS and Inverter<br>and associated<br>equipments   |
|                                                       | <b>Module 4</b><br>Solar Panel<br>Installation and<br>Service      | 0                | 250                    | OB 4.1<br>To understand the VI chara of solar cell and applications."<br>To understand the construction of Solar Panel<br>Learn the steps involved in installation of Solar Panel<br>Learn the steps involved in trouble shooting of Solar Panel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MO 4.1<br>To understand the VI chara of solar cell and applications."<br>To understand the construction of Solar Panel<br>Learn the steps involved in installation of Solar Panel<br>Learn the steps involved in trouble shooting of Solar Panel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | - Demonstration<br>- Practical Exercises |                                                    |

| COURSE NAME<br>&<br>CODE       | TOPIC /<br>MODULE          | THEOR<br>Y (Hrs) | PRACTI<br>CAL<br>(Hrs) | OBJECTIVE OF MODULE                                                         | OUTCOME OF MODULE                                                  | METHEDOLOGY                           | TOOLS<br>REQUIRED |
|--------------------------------|----------------------------|------------------|------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------|-------------------|
| <b>Project Work<br/>(2008)</b> | 1.Students Project<br>Work | 0                | 160                    | OB 1.1 To be familiar with industrial environment and<br>production process | Employ skills acquired to solve problems of social significance or | - Demonstration<br>- Industrial Visit |                   |